



2807 West County Road 75
Monticello, MN 55362

May 30, 2025

L-MT-25-022
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Subsequent Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant Licensee Event Report 2024-002-01

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-263/2024-002-01 per 10 CFR 50.73(a)(2)(i)(B), 10 CFR 50.73(a)(2)(v)(B), and 10 CFR 50.73(a)(2)(v)(D). This report is a supplement to LER 50-263/2024-002-00 which was submitted on August 27, 2024 (ADAMS accession number ML24240A166).

If you have any questions about this submittal, please contact Carrie Seipp, Senior Regulatory Engineer, at 612-330-5576.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Greg D. Brown'.

Greg D. Brown
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-263/2024-002-01**

4 pages follow



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name

Monticello Nuclear Generating Plant

☒ 050☐ 052

2. Docket Number

263

3. Page

1 OF 4

4. Title

Low Pressure Coolant Injection Inoperable Due To Motor Valve Failure

5. Event Date

6. LER Number

7. Report Date

8. Other Facilities Involved

Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
06	28	24	2024	- 002 -	01	05	30	25	Facility Name	☐ 052	Docket Number

9. Operating Mode

1

10. Power Level

100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☒ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☒ 50.73(a)(2)(i)(B)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Carrie Seipp, Senior Nuclear Regulatory Engineer

Phone Number (Include area code)

612-330-5576

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	BO	ISV	L200	Y					

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month Day Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 28, 2024 at 0110 CDT, Monticello Nuclear Generating Plant was in Mode 1 at 100 percent power and performing OSP-RHR-0556 "Residual Heat Removal (RHR) Water Fill Verification" Technical Specification Surveillance Procedure. When the RHR Low Pressure Coolant Injection (LPCI) Motor Valve MO-2012 "RHR Division 1 LPCI Injection Outboard Valve" was attempted to be cycled, the Outboard Motor Valve opened approximately one inch then stopped. With the Outboard Motor Valve incapable of opening, the A LPCI injection path was inoperable for injection as an Emergency Core Cooling system; and due to the plant design of the LPCI Loop Select Logic, this rendered both subsystems of LPCI inoperable. Subsequent evaluation determined that the Outboard Motor Valve was considered inoperable when it was configured closed based on the potential to have Reactor pressure on the inboard side of the valve due to leakage past the associated RHR Division 1 LCPI Injection Inboard Motor Valve.

The cause of the event was that the Outboard Motor Valve's motor pinion was not secured to the motor shaft because an incorrect-sized set screw was installed. The shorter motor pinion set screw relaxed, allowing the motor pinion to move axially along the motor shaft until it no longer engaged the worm shaft clutch gear.

On June 29, 2024, the Outboard Motor Valve's motor pinion was adjusted back into position on the motor shaft. On August 7, 2024, the incorrect motor pinion set screw on the Outboard Motor Valve was replaced with the correct set screw.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME	☒ 050 ☐ 052	2. DOCKET NUMBER	3. LER NUMBER		
Monticello Nuclear Generating Plant		263	YEAR 2024	SEQUENTIAL NUMBER 002	REV NO. 01

NARRATIVE**EVENT DESCRIPTION**

On June 28, 2024 at 0110 CDT, Monticello Nuclear Generating Plant (MNGP) was in Mode 1 at 100 percent power and performing OSP-RHR-0556 "Residual Heat Removal (RHR) Water Fill Verification" Technical Specification Surveillance Procedure. When the RHR Low Pressure Coolant Injection (LPCI) [EIS CODE: BO] Motor Valve MO-2012 "RHR Division 1 LPCI Injection Outboard Valve" was attempted to be opened via the control room hand switch, the Outboard Motor Valve opened approximately one inch then stopped. With the Outboard Motor Valve incapable of opening, the A LPCI injection path was inoperable for injection as an Emergency Core Cooling system; and due to the plant design of the LPCI Loop Select Logic, this rendered both subsystems of LPCI inoperable. Specifically, if the B Recirc loop was determined to be broken, the automatic logic would be incapable of opening the path to the A LPCI injection path and neither division would automatically inject. Subsequent evaluation determined that the Outboard Motor Valve was inoperable when it was configured closed based on the potential to have Reactor pressure on the inboard side of the valve due to leakage past the associated Inboard Motor Valve MO-2014 "RHR Division 1 LCPI Injection Inboard valve."

This event is reportable per 10 CFR 50.73(a)(2)(i)(B) for a condition prohibited by Technical Specifications and 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(D) for an event or condition that could have prevented the fulfillment of a safety function of structures or systems that are needed to remove residual heat and mitigate the consequences of an accident based on the safety function at the RHR LPCI system level. This issue was evaluated for a Safety System Functional Failure (SSFF) in accordance with NEI 99-02. This event is a Safety System Functional Failure (SSFF) due to meeting reporting criteria per 10 CFR 50.73(a)(2)(v)(B) and 10 CFR 50.73(a)(2)(v)(D).

DATES AND APPROXIMATE TIMES OF MAJOR OCCURANCES

- October 18, 2023 at 1407 CDT: The Outboard Motor Valve was closed due to leakage past the Inboard Motor Valve.
- October 22, 2023 at 0207 CDT: The required completion time to be in Mode 3 due to Technical Specification 3.5.1 Condition F, as directed by Condition D.
- November 15, 2023 at 1619 CST: The Outboard Motor Valve was opened.
- April 14, 2024 at 1455 CDT: The Outboard Motor Valve was closed due to leakage past the Inboard Motor Valve.
- April 18, 2024 at 0255 CDT: The required completion time to be in Mode 3 due to Technical Specification 3.5.1 Condition F, as directed by Condition D.
- June 27, 2024 at 2158 CDT: Technical Specification 3.5.1 Condition D entered for Technical Specification Surveillance Procedure OSP-RHR-0556.
- June 28, 2024 at 0110 CDT: Event Date. The Outboard Motor Valve was incapable of fully opening during Technical Specification Surveillance Procedure SP-RHR-0556.
- June 29, 2024 at 1533 CDT: Technical Specification 3.5.1 Condition D exited following completion of maintenance on the Outboard Motor Valve.
- August 05, 2024: Identification that the adjustment made to the Outboard Motor Valve's motor pinion, completed on June 29, 2024, should not have been possible if the motor pinion had been installed per the vendor technical manual.
- August 07, 2024 at 0500 CDT: Technical Specification 3.5.1 Condition B and D entered for investigation of and repairs to the Outboard Motor Valve.
- August 08, 2024 at 1058 CDT: Technical Specification 3.5.1 Condition B and D exited following installation of the correct motor pinion set screw on the Outboard Motor Valve.
- April 03, 2025: Determination that the Outboard Motor Valve should have been considered inoperable when it was configured closed with the incorrect motor pinion set screw and had the potential to have Reactor pressure on the inboard side of the valve due to leakage past the Inboard Motor Valve.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER		3. LER NUMBER		
Monticello Nuclear Generating Plant		263		YEAR	SEQUENTIAL NUMBER	REV NO.
				2024	002	01

NARRATIVE**EVENT ANALYSIS**

On June 27, 2024 at 2158 CDT, Technical Specification 3.5.1 "ECCS - Operation" Condition D "Two LPCI subsystems inoperable for reasons other than Condition C and G" was entered for performance of Technical Specification Surveillance Procedure 0255-04-IA-1-2 "RHR Loop B Quarterly Pump and Valve Tests." Prior to exiting Condition D of Technical Specification 3.5.1, OSP-RHR-0556 was performed. On June 28, 2024 at 0110 CDT while the Outboard Motor Valve was attempted to be opened via the control room hand switch, the Outboard Motor Valve opened approximately one inch then stopped, resulting in the inability to exit Condition D of Technical Specification 3.5.1. The plant had no other Structures, Systems, or Components that were inoperable at the start of the event that contributed to the event.

While this was later determined to be incorrect, initial troubleshooting identified that the most likely cause of the Outboard Motor Valve failure to stroke was the clutch mechanism was disengaged partially from the motor pinion as a result of foreign material from failure of the tripper spring. The Outboard Motor Valve was repaired and satisfactorily retested, then Condition D of Technical Specification 3.5.1 was exited on June 29, 2024 at 1533 CDT.

During performance of the root cause evaluation interviews, it was identified that during the Outboard Motor Valve repairs performed on June 29, 2024, the motor pinion was found extended on the motor shaft and prior to reassembly, it was adjusted back into position. The vendor technical manual specified that the motor pinion should have a set screw and lock wire installed, so it should not have been able to move.

Further physical investigation on the Outboard Motor Valve and specifically the motor pinion was performed on August 7, 2024. The Outboard Motor Valve was successfully stroked open and closed, then the motor pinion was found to be held in place with a set screw and lock wire, with no movement along the motor shaft. The set screw was removed for inspection, and it was determined to be the incorrect set screw. Additionally, indications on the motor shaft revealed that motor pinion had potentially moved axially in the past. A review of past maintenance records did not identify a time where the set screw had been replaced.

An incorrect set screw would have allowed the motor pinion to travel axially along the motor shaft and eventually disengage with the worm shaft clutch gear. When the motor pinion disengaged with the worm shaft clutch gear, the Outboard Motor Valve could not be operated via the control room hand switch which starts the motor actuator.

On April 03, 2025, the determination was made that the Outboard Motor Valve should have been considered inoperable when it was configured closed with the incorrect motor pinion set screw and had the potential to have Reactor pressure on the inboard side of the valve due to leakage past the Inboard Motor Valve. With the Outboard Motor Valve incapable of opening, the A LPCI injection path was inoperable for injection as an Emergency Core Cooling system; and due to the plant design of the LPCI Loop Select Logic, this rendered both subsystems of LPCI inoperable. Technical Specification 3.5.1 Condition D requires restoration of one LPCI subsystem to Operable status within 72 hours. When Condition D is not met, Condition F requires entry to Mode 3 within 12 hours. Therefore, the plant was in a condition prohibited by Technical specifications in the following time frames:

- October 22, 2023 at 0207 CDT through November 15, 2023 at 1611 CST.
- April 18, 2024 0255 CDT through June 29, 2024 at 1533 CDT.

The plant was in a condition that could have prevented the fulfillment of a safety function in the following timeframes:

- October 18, 2023 at 1407 CDT through November 15, 2023 at 1611 CST.
- April 14, 2024 at 1455 CDT through June 29, 2024 at 1533 CDT.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER		
Monticello Nuclear Generating Plant	☒ 050	263	YEAR	SEQUENTIAL NUMBER	REV NO.
	☐ 052		2024	002	01

NARRATIVE**ASSESSMENT OF SAFETY CONSEQUENCES**

Actual safety consequences of the event were very low for both normal operation and design basis event operation. High Pressure Emergency Core Cooling System (ECCS), the Low Pressure ECCS subsystem of Core Spray, and the remaining functions of RHR including Drywell Spray and Suppression Pool Cooling were not affected. A probabilistic risk assessment was performed to estimate the impact of the unavailability of the Outboard Motor Valve on Core Damage Frequency and Large Early Release Frequency. The results were within the bounds of "Very Low Safety Significance" described in NRC Inspection Manual Chapter 0609. There were no radiological or industrial impacts associated with this event. The health and safety of the public and site personnel were not impacted during this event.

CAUSE OF THE EVENT

The Outboard Motor Valve failed to operate because the motor pinion was not secured to the motor shaft because an incorrect-sized set screw was installed. The shorter motor pinion set screw relaxed, allowing the motor pinion to move axially along the motor shaft until it no longer engaged the worm shaft clutch gear.

CORRECTIVE ACTIONS

On June 29, 2024, the Outboard Motor Valve's motor pinion was adjusted back into position on the motor shaft. On August 7, 2024, the incorrect motor pinion set screw on the Outboard Motor Valve was replaced with the correct set screw.

PREVIOUS SIMILAR EVENTS

There were no previous similar events in the past three years.